

Bill of Materials — Internal Draft

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User Request

@quotepilot-bom

We're quoting a mid-size corporate office expansion for a client adding 150 desks to their existing campus. They need 24-port managed PoE switches for the new floor, a next-generation firewall to handle increased perimeter traffic, and Cat6A structured cabling to support the buildout. Budget is flexible but the client prefers not to overspend on switching. Please generate a full 3-tier BOM.

DATA FRESHNESS WARNINGS

- VEN-002 (Juniper Networks): effective_date 2026-02-18 is 82d old, exceeding the 60d threshold (rule-002). Budget switch V002-SW-0001 is included per policy; SE must acknowledge before customer delivery.
- VEN-004 (Arista Networks): effective_date 2026-02-24 is 76d old (rule-002). SKU V004-SW-0003 not selected for any tier but appears in catalog pool.
- VEN-006 (Extreme Networks): effective_date 2025-11-02 is 190d old (rule-002). No SKUs from this vendor selected; catalog pool entry noted.
- VEN-008 (Fortinet): effective_date 2026-02-15 is 85d old (rule-002). No SKUs selected.
- VEN-012 (SonicWall): effective_date 2025-06-05 is 340d old (rule-002). No SKUs selected; extremely stale — catalog team should be alerted.
- VEN-022 (CablePro Supply): effective_date 2025-08-22 is 262d old (rule-002). No SKUs selected; very stale.
- VEN-027 (TD Synnex): effective_date 2026-03-01 is 71d old (rule-002). No SKUs selected for chosen tiers.
- VEN-028 (Ingram Micro): effective_date 2026-03-07 is 65d old (rule-002). No SKUs selected.
- VEN-030 (SHI International): effective_date 2026-02-18 is 82d old (rule-002). No SKUs selected.
- VEN-031 (Lenovo Networking): effective_date 2026-02-27 is 73d old (rule-002). No SKUs selected.
- VEN-034 (Sophos): effective_date 2026-02-18 is 82d old (rule-002). No SKUs selected.
- VEN-036 (Trellix/FireEye): effective_date 2026-03-01 is 71d old (rule-002). No SKUs selected.
- VEN-042 (Tripp Lite): effective_date 2026-02-24 is 76d old (rule-002). No SKUs selected.
- VEN-046 (Axis Communications): effective_date 2026-02-21 is 79d old (rule-002). No SKUs selected for active tiers.

RECOMMENDATION

For the 150-desk campus expansion, the recommended configuration is: balanced switching (V043-SW-0001, \$1,401.03/unit) to honor the client's preference not to overspend, premium NGFW (V009-SEC-0001, \$10,472.55) to ensure the enterprise-edge throughput (6-15Gbps) needed for the expanded perimeter, and balanced structured cabling (V021-CAB-0001, \$473.79/panel) as a clean, name-brand infrastructure anchor. This cross-tier configuration delivers a blended margin of approximately 22.4% across the three anchor line items. The only advisory to resolve before delivery is the stale-data warning on the budget switching alternative (Juniper Networks, VEN-002, 82d since last price confirmation) — the balanced and premium switch picks carry no such flags and are ready to present.

Tiered Recommendations

BUDGET TIER							
SKU	Vendor	Category	Availability	Lead Time	Unit Cost	Margin	Effective Date
V002-SW-0001	Juniper Networks VEN-002	networking-switch	In Stock	10 days	\$1055.74	16.2%	2026-02-18
Evidence: catalog: VEN-002/V002-SW-0001 effective_date 2026-02-18 · ruleset: rule-002 (stale: 82d since effective_date, threshold 60d — included per policy; SE review required) · ruleset: rule-001 (lead_time 10d — within 21d threshold, no breach) · ruleset: rule-003 (availability: true — no gate triggered) · ruleset: rule-004 (margin 16.2% — above 15.0% floor, no flag)							
V009-SEC-0002	Check Point Software VEN-009	network-security	In Stock	21 days	\$666.22	27.3%	2026-04-26
Evidence: catalog: VEN-009/V009-SEC-0002 effective_date 2026-04-26 · ruleset: rule-001 (lead_time 21d — at threshold boundary, no breach) · ruleset: rule-002 (effective_date 15d ago — within 60d freshness window, no warning) · ruleset: rule-003 (availability: true — no gate triggered) · ruleset: rule-004 (margin 27.3% — above 15.0% floor, no flag)							
V020-CAB-0004	Belden VEN-020	structured-cabling	In Stock	2 days	\$457.89	20.4%	2026-05-01
Evidence: catalog: VEN-020/V020-CAB-0004 effective_date 2026-05-01 · ruleset: rule-001 (lead_time 2d — well within 21d threshold, no breach) · ruleset: rule-002 (effective_date 10d ago — within 60d freshness window, no warning) · ruleset: rule-003 (availability: true — no gate triggered) · ruleset: rule-004 (margin 20.4% — above 15.0% floor, no flag)							

BALANCED TIER

SKU	Vendor	Category	Availability	Lead Time	Unit Cost	Margin	Effective Date
V043-SW-0001	APC by Schneider VEN-043	networking-switch	In Stock	14 days	\$1401.03	18.4%	2026-04-14
Evidence: catalog: VEN-043/V043-SW-0001 effective_date 2026-04-14 · ruleset: rule-001 (lead_time 14d — within 21d threshold, no breach) · ruleset: rule-002 (effective_date 27d ago — within 60d freshness window, no warning) · ruleset: rule-003 (availability: true — no gate triggered) · ruleset: rule-004 (margin 18.4% — above 15.0% floor, no flag)							
V041-SEC-0002	Black Box Corporation VEN-041	network-security	In Stock	21 days	\$5231.05	22.1%	2026-03-31
Evidence: catalog: VEN-041/V041-SEC-0002 effective_date 2026-03-31 · ruleset: rule-001 (lead_time 21d — at threshold boundary, no breach) · ruleset: rule-002 (effective_date 41d ago — within 60d freshness window, no warning) · ruleset: rule-003 (availability: true — no gate triggered) · ruleset: rule-004 (margin 22.1% — above 15.0% floor, no flag)							
V021-CAB-0001	Legrand VEN-021	structured-cabling	In Stock	3 days	\$473.79	18.4%	2026-03-25
Evidence: catalog: VEN-021/V021-CAB-0001 effective_date 2026-03-25 · ruleset: rule-001 (lead_time 3d — well within 21d threshold, no breach) · ruleset: rule-002 (effective_date 47d ago — within 60d freshness window, no warning) · ruleset: rule-003 (availability: true — no gate triggered) · ruleset: rule-004 (margin 18.4% — above 15.0% floor, no flag)							

PREMIUM TIER

SKU	Vendor	Category	Availability	Lead Time	Unit Cost	Margin	Effective Date
V041-SW-0001	Black Box Corporation VEN-041	networking-switch	In Stock	2 days	\$1606.72	21.0%	2026-03-31
Evidence: catalog: VEN-041/V041-SW-0001 effective_date 2026-03-31 · ruleset: rule-001 (lead_time 2d — well within 21d threshold, no breach) · ruleset: rule-002 (effective_date 41d ago — within 60d freshness window, no warning) · ruleset: rule-003 (availability: true — no gate triggered) · ruleset: rule-004 (margin 21.0% — above 15.0% floor, no flag)							
V009-SEC-0001	Check Point Software VEN-009	network-security	In Stock	4 days	\$10472.55	28.2%	2026-04-26
Evidence: catalog: VEN-009/V009-SEC-0001 effective_date 2026-04-26 · ruleset: rule-001 (lead_time 4d — well within 21d threshold, no breach) · ruleset: rule-002 (effective_date 15d ago — within 60d freshness window, no warning) · ruleset: rule-003 (availability: true — no gate triggered) · ruleset: rule-004 (margin 28.2% — above 15.0% floor, no flag)							
V024-CAB-0002	Vertiv VEN-024	structured-cabling	In Stock	3 days	\$682.11	20.5%	2026-03-28
Evidence: catalog: VEN-024/V024-CAB-0002 effective_date 2026-03-28 · ruleset: rule-001 (lead_time 3d — well within 21d threshold, no breach) · ruleset: rule-002 (effective_date 44d ago — within 60d freshness window, no warning) · ruleset: rule-003 (availability: true — no gate triggered) · ruleset: rule-004 (margin 20.5% — above 15.0% floor, no flag)							

Justification

Budget Tier

Budget tiers minimize unit cost while meeting all rule thresholds. The switching budget pick carries a stale-data warning from Juniper Networks that the SE must acknowledge before finalizing.

- networking-switch: VEN-002/V002-SW-0001 (Juniper Networks) at \$1,055.74 — lowest-cost qualifying 24-port PoE+ managed switch in the filtered catalog
- Lead-time compliance (rule-001): 10d — within the 21d threshold; no substitution required
- Margin compliance (rule-004): 16.2% — above the 15.0% floor; no flag triggered
- rule-002 flag: Juniper Networks effective_date 2026-02-18 is 82d old, exceeding the 60d freshness threshold — requires SE acknowledgment before customer delivery
- network-security: VEN-009/V009-SEC-0002 (Check Point Software) at \$666.22 — branch-tier NGFW (1-5Gbps), lowest-cost NGFW candidate with no rules fired and 27.3% margin
- structured-cabling: VEN-020/V020-CAB-0004 (Belden) at \$457.89 — lowest-cost clean 48-port Cat6A patch panel, 2d lead time, 20.4% margin

Balanced Tier

Balanced tiers select mid-cost, rule-clean SKUs that align with the SE's guidance that the client prefers not to overspend on switching. All three balanced line items carry no rule flags.

- networking-switch: VEN-043/V043-SW-0001 (APC by Schneider) at \$1,401.03 — clean 24-port PoE+ managed switch, 14d lead time, aligned with cost-conscious switching direction
- Lead-time compliance (rule-001): 14d — within the 21d threshold; no substitution required

- Margin compliance (rule-004): 18.4% — above the 15.0% floor; no flag triggered
- No rule-002, rule-003, or rule-004 flags on any balanced line item across all three categories
- network-security: VEN-041/V041-SEC-0002 (Black Box Corporation) at \$5,231.05 — enterprise-edge NGFW (6-15Gbps) stepping up from branch-tier budget; appropriate for 150-desk perimeter expansion
- structured-cabling: VEN-021/V021-CAB-0001 (Legrand) at \$473.79 — established structured-cabling brand, 3d lead time; trade-off vs. budget is a modest \$15.90 premium for Legrand's enterprise pedigree

Premium Tier

Premium tiers maximize performance and delivery speed. The switching premium remains cost-disciplined at \$1,606.72 — only 14.7% above the balanced pick — reflecting the SE's guidance to avoid over-investment in switching.

- networking-switch: VEN-041/V041-SW-0001 (Black Box Corporation) at \$1,606.72 — fastest lead time of all clean 24-port PoE+ candidates at 2d, 21.0% margin
- Lead-time compliance (rule-001): 2d — well within the 21d threshold; no substitution required
- Margin compliance (rule-004): 21.0% — above the 15.0% floor; no flag triggered
- No rule flags on any premium line item; all three premium SKUs are fully clean
- network-security: VEN-009/V009-SEC-0001 (Check Point Software) at \$10,472.55 — enterprise-edge NGFW (6-15Gbps) with highest margin of clean enterprise candidates at 28.2% and 4d lead time
- structured-cabling: VEN-024/V024-CAB-0002 (Vertiv) at \$682.11 — highest-cost clean patch panel with best margin (20.5%) and 3d lead time; trade-off vs. balanced is a \$208.32 step-up for Vertiv's data-center-grade pedigree

Status: **Requires SE review before customer delivery**